

POWELL RIVER, BC, Canada

WMO: 717200

Lat: 49.8344N

Lon: 124.4981W

Elev: 410

StdP: 14.48

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.8	25.6	12.7	10.7	26.8	19.2	14.6	29.5	25.2	44.3	22.5	43.2	2.0	90	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	80.5	64.2	77.1	62.9	74.0	61.8	65.7	77.0	64.2	74.4	62.8	71.7	6.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	82.3	69.4	60.1	78.8	68.0	58.9	75.4	66.7	30.6	77.5	29.5	74.7	28.5	71.8	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	17.1	14.5	DB	16.2	86.8	3.5	4.0	13.7	89.6	11.6	91.9	9.6	94.1	7.1	97.0	(4)
(5)			WB	15.1	68.7	3.5	1.9	12.6	70.1	10.6	71.2	8.6	72.3	6.1	73.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	38.1	37.8	41.5	46.0	54.0	58.6	63.6	63.4	57.2	48.4	41.7	36.8
	DBStd	10.80	5.59	5.53	4.45	4.23	4.89	4.61	4.02	3.89	4.66	4.35	5.80	5.49
	HDD50	1873	370	342	264	133	18	1	0	0	2	82	253	408
	HDD65	5910	835	762	728	570	341	200	72	77	238	515	699	873
	CDD50	1508	0	0	1	12	142	259	423	417	217	32	5	0
	CDD65	71	0	0	0	0	0	9	30	29	3	0	0	0
	CDH74	700	0	0	0	0	19	105	284	250	42	0	0	0
	CDH80	121	0	0	0	0	0	19	57	41	4	0	0	0
Wind	WSAvg	5.5	5.4	4.9	6.2	6.0	5.3	5.6	5.6	4.9	5.0	5.6	6.2	5.6
Precipitation	PrecAvg	44.7	5.7	4.1	3.9	2.5	2.6	2.1	1.8	2.0	2.4	5.2	6.4	6.3
	PrecMax	56.5	11.7	8.6	7.3	4.8	4.8	4.7	4.1	7.2	6.3	10.6	10.7	9.1
	PrecMin	27.0	1.3	1.7	1.8	0.5	0.9	0.4	0.0	0.1	0.2	1.1	2.4	1.7
	PrecStd	6.7	2.6	1.7	1.5	1.1	1.1	1.1	1.0	1.6	1.5	2.4	2.4	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.4	51.3	59.4	66.7	76.9	82.8	86.0	84.5	79.1	63.6	56.4	51.4
		MCWB	49.3	48.2	49.2	54.5	58.5	63.2	66.7	65.3	63.3	56.4	53.1	49.4
	2%	DB	49.1	49.3	53.6	60.0	71.9	76.7	80.4	80.1	73.3	59.9	53.8	47.9
		MCWB	47.0	46.5	47.7	50.0	58.3	61.6	64.4	64.5	61.2	55.2	51.2	45.9
	5%	DB	47.6	47.4	50.8	56.7	68.0	71.8	76.8	76.5	69.1	57.6	51.7	46.3
		MCWB	45.6	45.0	46.1	48.4	57.2	59.7	63.2	63.9	59.8	53.7	49.7	44.5
	10%	DB	45.8	45.5	48.7	54.0	64.4	67.9	73.6	73.0	65.8	55.5	49.7	44.7
		MCWB	44.0	43.4	45.1	47.1	55.3	57.7	62.1	62.3	58.4	52.3	47.7	43.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	49.9	51.5	55.6	61.1	65.6	68.6	67.5	65.4	58.6	54.5	49.9
		MCDB	50.9	50.7	55.0	63.3	72.4	78.1	84.3	79.5	75.7	60.4	56.1	51.1
	2%	WB	47.5	47.5	48.8	52.3	59.1	62.8	65.8	65.6	62.7	56.7	51.7	46.6
		MCDB	48.4	48.5	52.3	57.8	68.9	74.7	77.1	76.9	70.2	59.0	53.1	47.6
	5%	WB	45.9	45.1	46.9	49.9	57.5	60.5	64.0	64.3	60.9	54.6	49.9	44.8
		MCDB	47.2	46.6	50.2	54.4	66.1	69.2	74.2	74.5	67.2	56.9	51.7	46.1
	10%	WB	44.2	43.7	45.5	48.3	56.0	58.8	62.6	62.9	59.2	52.4	48.0	43.1
		MCDB	45.6	45.3	48.3	52.2	63.2	66.1	71.8	71.2	64.2	54.8	49.6	44.5
Mean Daily Temperature Range	5% DB	MDBR	7.9	10.0	12.2	14.6	17.1	15.8	18.2	18.1	15.3	11.4	9.3	7.5
		MCDBR	8.5	11.2	15.5	20.5	25.2	24.2	24.9	24.8	22.2	14.4	8.4	6.8
		MCWBR	7.0	8.1	9.9	11.8	12.9	11.0	11.2	11.4	11.3	9.6	6.8	5.7
	5% WB	MCDBR	8.0	8.7	13.6	15.6	21.8	20.7	22.8	22.5	18.8	11.3	7.7	6.8
MCWBR		6.8	6.7	9.3	9.3	9.7	11.5	10.1	10.6	10.9	10.0	8.1	6.5	6.0
Clear-Sky Solar Irradiance	taub	0.314	0.308	0.333	0.353	0.374	0.373	0.369	0.369	0.340	0.337	0.317	0.307	
	taud	2.421	2.510	2.447	2.414	2.397	2.424	2.470	2.473	2.557	2.524	2.491	2.410	
	Ebn at Noon	232	268	278	283	281	281	280	275	274	253	231	216	
	Edh at Noon	20	23	29	34	36	35	33	31	26	23	19	18	
All-Sky Solar Radiation	RadAvg	253	530	788	1233	1615	1739	1914	1648	1146	613	299	198	
	RadStd	38	89	109	129	177	197	206	138	102	105	58	30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page